

resources and local environments if not managed carefully.”

Cooling systems are another problem. Conventional hyperscale campuses can consume millions of litres of water daily. In a country where several urban centres already face chronic water shortages, this creates the possibility of technology infrastructure competing with households and agriculture for basic resources. An unchecked AI infrastructure boom could create a paradox where the technology meant to optimise human systems ends up stressing the very resources people depend upon.

“Amazon does not use water for cooling its data centres in India. In 2024, Amazon announced a goal to return more water to communities in India than it uses in its direct operations by 2027,” says Saji PK, Director, AWS data centre operations, Asia Pacific, Japan and China. According to him, at AWS, water conservation has become central to facility design in India.

AWS has invested over ₹42 crore in water replenishment projects and expects its investments to restore more than three billion litres of water annually. Its facilities increasingly rely on air-cooling systems along with liquid-cooling capabilities for advanced AI chips.

Microsoft’s India facilities (at Hyderabad, Pune, Chennai and Mumbai) are being engineered for zero-water cooling while relying on renewable energy partnerships with companies such as ReNew and Amplus Solar.

According to the International Energy Agency (IEA), a 100 MW AI data centre will consume around two million litres of water daily, about as much water consumed by 6,500 households in a day. Presently, only 25% of the water requirement globally is from recycled sources or non-potable water



HIMANI AGRAWAL

COO, MICROSOFT
INDIA AND SOUTH ASIA

“India’s digital public infrastructure has created one of the strongest foundations for innovation globally. The next opportunity is to build on that momentum by enabling AI infrastructure at scale”



VIBHUTI GARG

DIRECTOR, SOUTH ASIA, INSTITUTE OF ENERGY ECONOMICS AND FINANCIAL ANALYSIS

“India’s data centre boom is a double-edged sword. It will underpin the digital economy, but it will also intensify pressure on electricity systems, water resources, local environments”

sources, and the rest is fresh water.

The location of these centres is another challenge. Mumbai accounts for 53% of India’s total data centre capacity, and they contribute to 5% of the city’s peak electricity demand. Chennai and the Delhi National Capital Region (NCR) were also sought-after locations because of better connectivity. But that is now changing as operators are scouting for locations based on power availability and renewable-energy potential rather than proximity to urban centres. While companies aim to strike a balance, AI data centres tend to select places where they can access massive, stable electric grids.

“Geographic diversification is key. Concentration in a few urban hubs (Mumbai, NCR) amplifies stress on already constrained water and power resources. Expanding into Tier II sites (like Visakhapatnam, Bhubaneswar)

can ease pressure on resources,” says Garg. Locating data centres in high renewable potential states such as Rajasthan or Tamil Nadu could reduce transmission losses and improve the reliability of green supply.

Developers could also look at sustainable models outside India. Climate-wise, India is like the United Arab Emirates (UAE) or Singapore, facing extreme heat and humidity. In humid coastal hubs like Mumbai, India could adopt Singapore’s tropical standards by raising operating temperatures to reduce cooling loads. In arid regions like Delhi, India could emulate Arizona and the UAE by mandating closed-loop liquid cooling and deploying dry air radiators.

The future of AI, increasingly, may depend not only on how intelligently machines think, but on how intelligently humans power them. **BT**